

Adil Shiyas

Education

adilshiyas@gmail.com | (508)-365-7029 | Worcester, MA | website | github

Worcester Polytechnic Institute

Master of Science in Robotics Engineering | GPA: 3.8

Aug 2024 - May 2026

Worcester, MA

PES University

Bachelor of Technology in Mechanical Engineering

Aug 2017 - June 2021

Bangalore, India

Experience

Graduate Researcher – ELPIS Lab, Worcester Polytechnic Institute

May 2025 – May 2026

Principal Investigator: Professor Constantinos Chamzas

Worcester, MA

- **COAD: Constant-Time Planning for Continuous Goal Regions** (GitHub) – Developed a motion planning framework for robotic manipulation under continuous goal regions, enabling constant-time trajectory retrieval through offline precomputation and online adaptation; validated in MuJoCo and on a real UR10 robot.
- **MotionBenchMaker-Pinocchio** (GitHub) – Ported MotionBenchMaker from ROS1 (C++) to a Python-based framework using Pinocchio, Coal, and OMPL, eliminating simulator overhead and enabling scalable benchmarking of motion planners across multiple manipulators and environments.

Research Associate – International Institute of Information Technology - Bangalore (IIIT-B)

Oct 2022 – June 2024

Principal Investigator: Professor Sachit Rao

Bangalore, India

- **Trochoidal Paths for a Multi-Robot Swarm** (GitHub) – Formulated geometric and temporal constraints on a consensus-based multi-robot controller and solved for controller parameters in MATLAB to generate collision-free trochoidal trajectories; validated on an indoor mobile robot platform and scaled to 12 simulated agents.
- **Reciprocal Collision Avoidance** – Developed a decentralized reciprocal collision avoidance controller based on collision cones, enabling reactive multi-robot navigation using local state information while mitigating deadlocks and oscillations.

Projects

Task and Motion Planning Framework (ROS 2, MoveIt 2, PDDL) (GitHub)

Mar 2026 – May 2026

- Developed a ROS 2 task and motion planning package integrating PDDL, Fast Downward, and MoveIt 2 to generate and execute symbolic manipulation plans on a Franka Emika Panda manipulator.
- Designed Python and C++ nodes using MoveIt Task Constructor, OMPL, and inverse kinematics to bridge symbolic task planning with geometric motion execution for the Tower of Hanoi problem.

Autonomous Driving Perception and Visualization System

Feb 2025 – Apr 2025

- Developed a Tesla-style perception pipeline using Detic, Mask R-CNN, Marigold, OpenPifPaf, and RAFT to estimate object semantics, lane geometry, depth, pose, and motion from monocular driving videos.
- Integrated Gemma 3 for traffic light state estimation and speed limit recognition.
- Constructed and rendered a 3D driving scene in Blender using fused perception and depth estimation outputs.

Action Chunking Transformer for Vision-Based Robotic Peg Insertion

Feb 2025 – April 2025

- Implemented an Action Chunking Transformer (ACT) for imitation learning of UR10 peg insertion, mapping multi-modal observations (RGB + state) to temporally consistent action sequences.
- Built a demonstration pipeline using scripted expert policies and randomized initial conditions; leveraged a transformer-VAE architecture to improve robustness over vanilla behavior cloning.

Skills

Programming: Python, C++17, MATLAB | **Robotics:** OMPL, MoveIt 2, ROS 2, Pinocchio | **Libraries:** PyTorch, OpenCV |

Simulation: MuJoCo, Genesis, PyBullet | **Tools & Systems:** Linux, Simulink, Blender, Fusion 360

Publications

COAD: Constant-Time Planning for Continuous Goal Manipulation with Compressed Library and Online Adaptation

Adil Shiyas, Zhuoyun Zhong, Constantinos Chamzas.

Under Review at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [arXiv]

Design of Planar Collision-Free Trochoidal Paths for a Multi-Robot Swarm

Adil Shiyas, Sachit Rao

European Journal of Control, Vol. 81, 2025. [ScienceDirect]

Reciprocal Collision Avoidance in Non-holonomic Robots using Collision Cones

Adil Shiyas, Sachit Rao

2025 Eleventh Indian Control Conference (ICC) [IEEEExplore]